

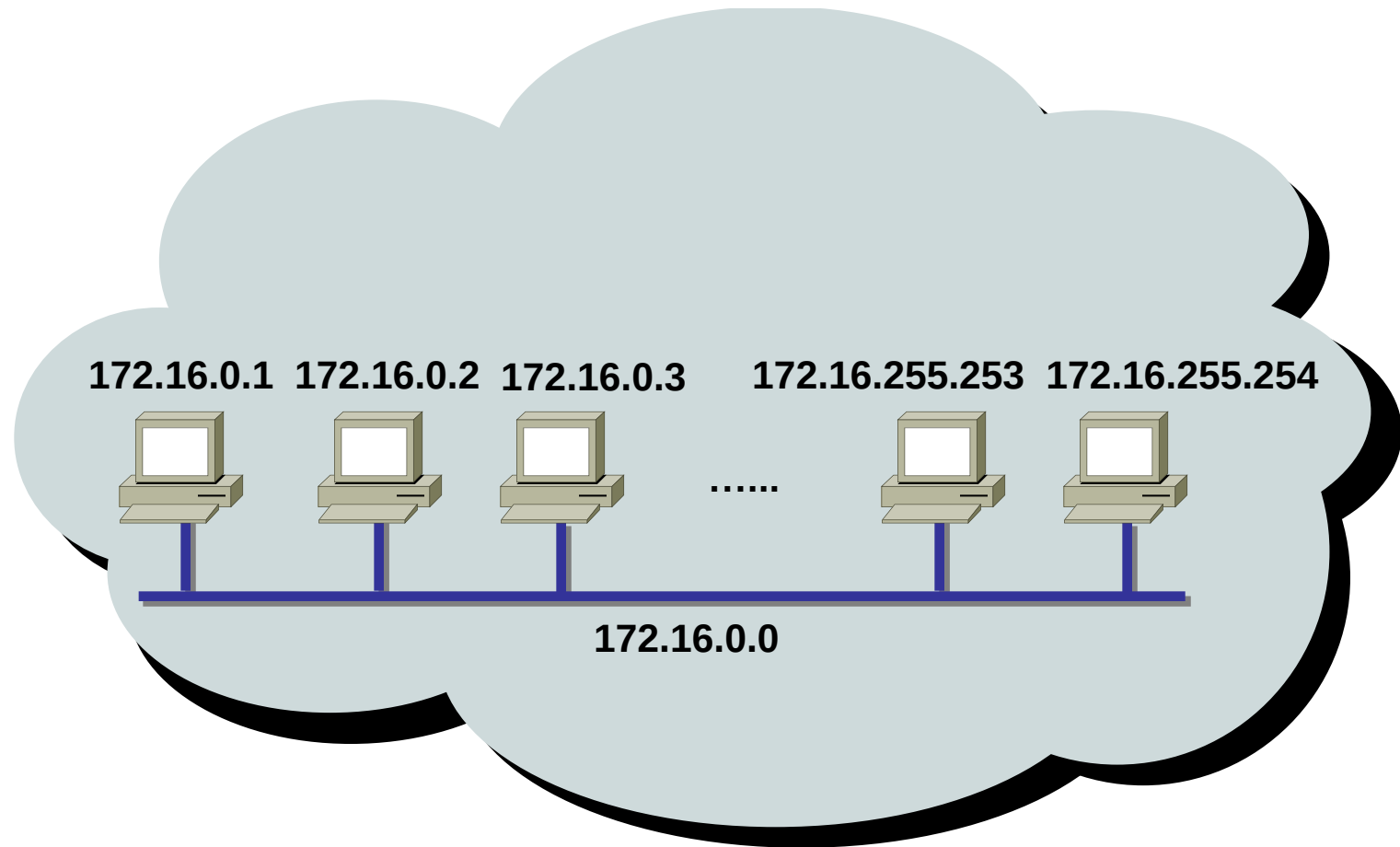


Computer Network

Chapter 3 : Introduction Subnetting



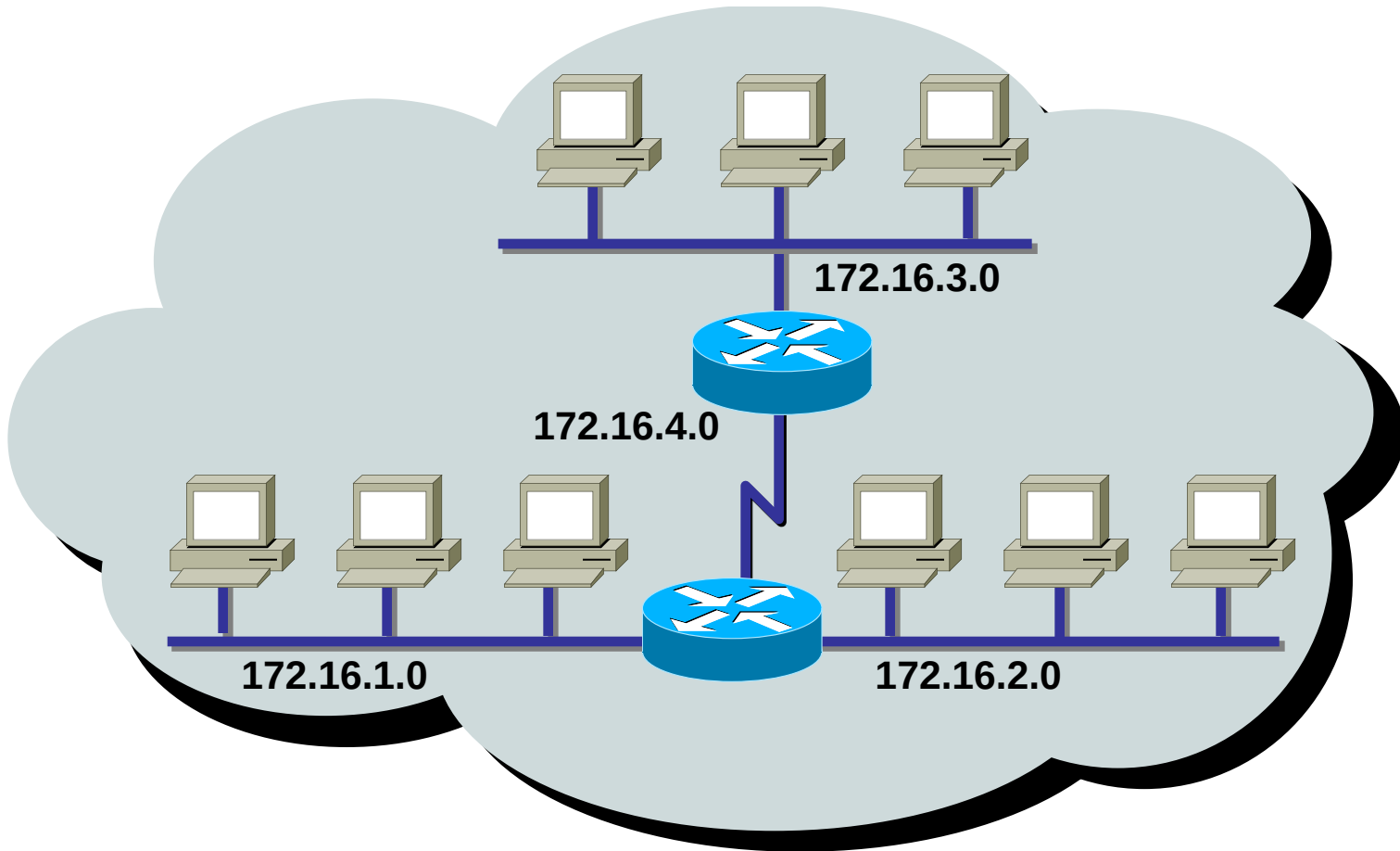
Addressing without subnet



- Network 172.16.0.0



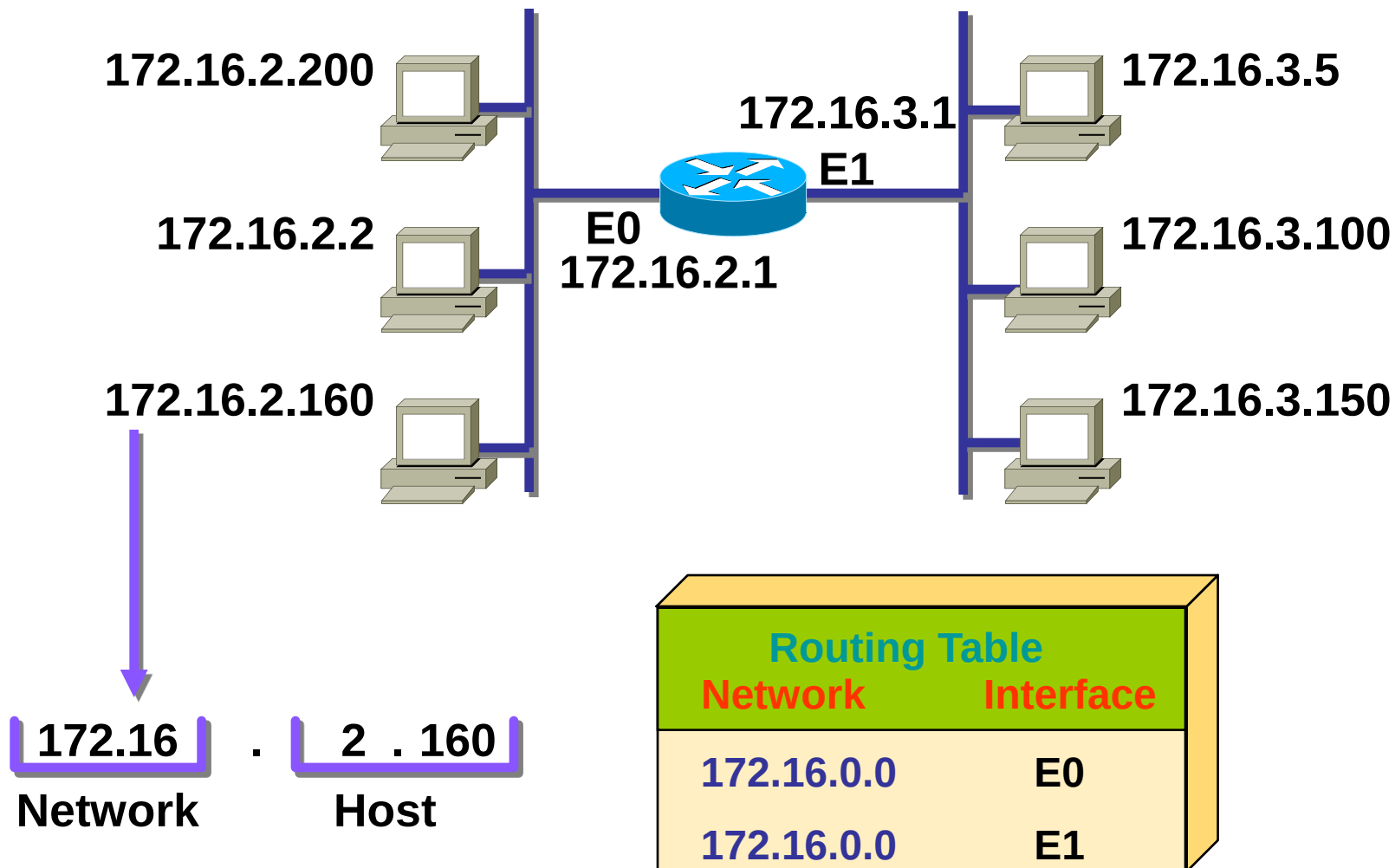
Addressing with subnet



Network 172.16.0.0

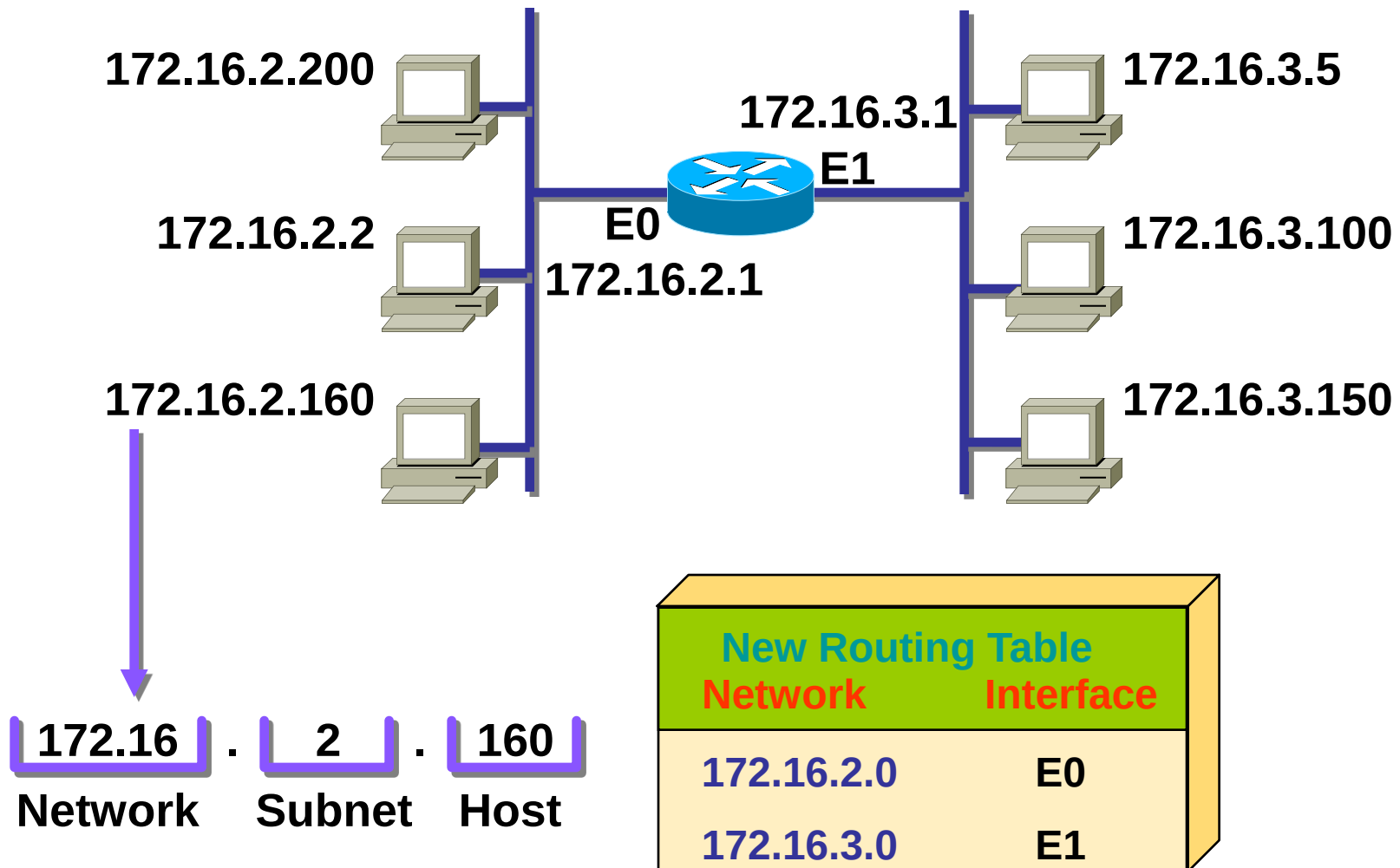


Addressing without subnet



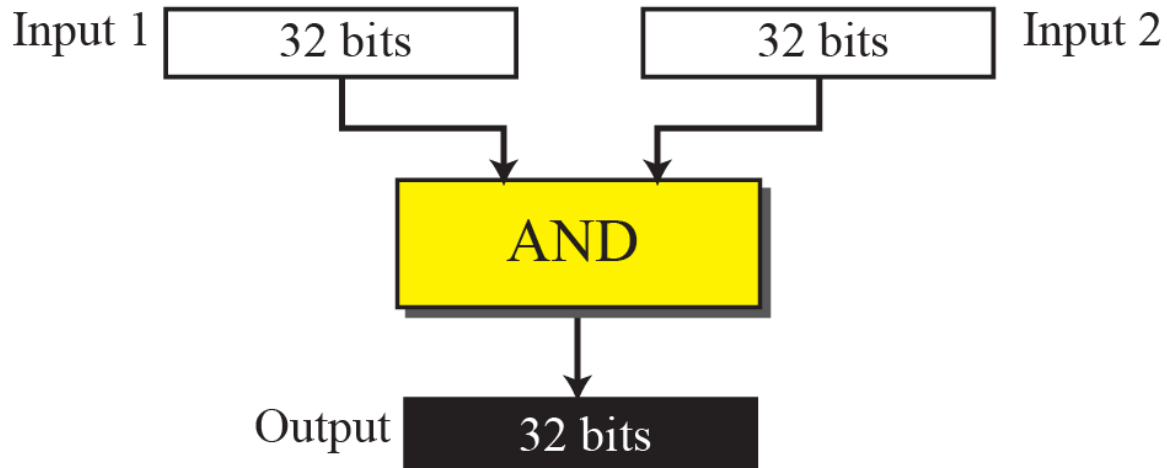


Addressing with subnet





Bitwise AND operation



AND

Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

Operation for each bit

$$X \text{ AND } 1 \rightarrow X$$



Default subnet mask

A **network mask** or a **default subnet mask** in classfull addressing is a 32-bit number with n leftmost bits all set to 1s and $(32 - n)$ rightmost bits all set to 0s (net-bits is all 1s, host-bits is all 0s).

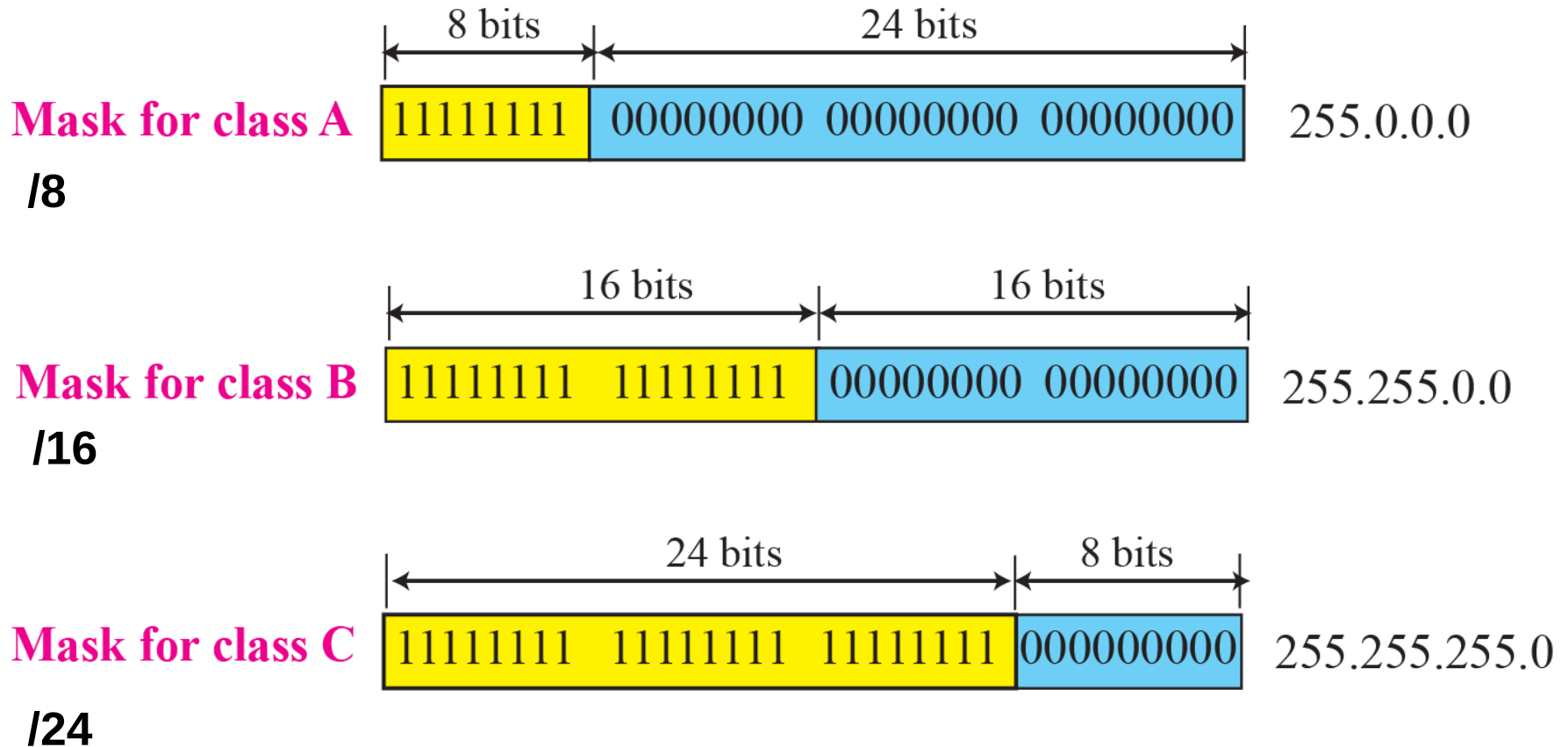
Can be written as **/n** with n is the number of bit 1.

Example: 255.255.255.0
or /24

→ Subnet mask for class A,B,C ??????

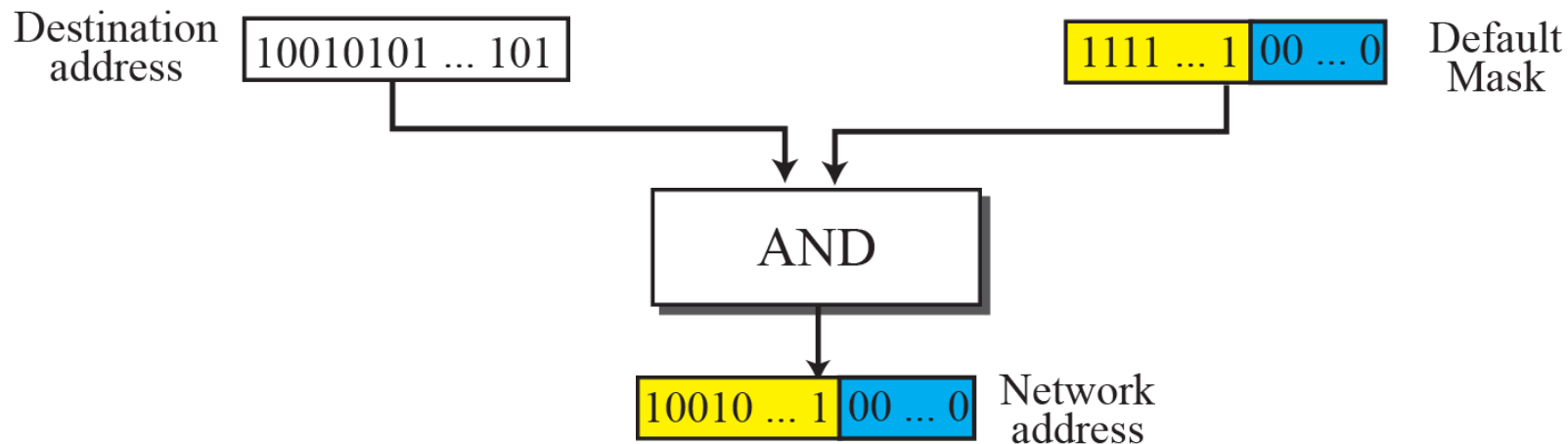


Default Subnet mask





Finding a network address using the default subnet mask



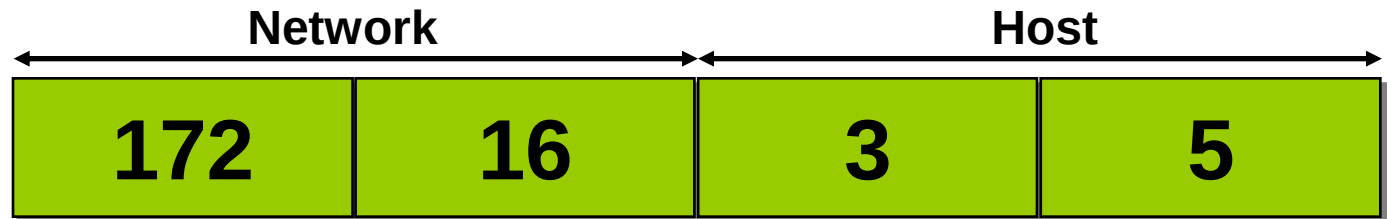
Example: find the network address of

- 192.168.10.15 - 255.255.255.0
- 10.1.5.6 - 255.0.0.0
- 172.10.9.10 - 255.255.255.0

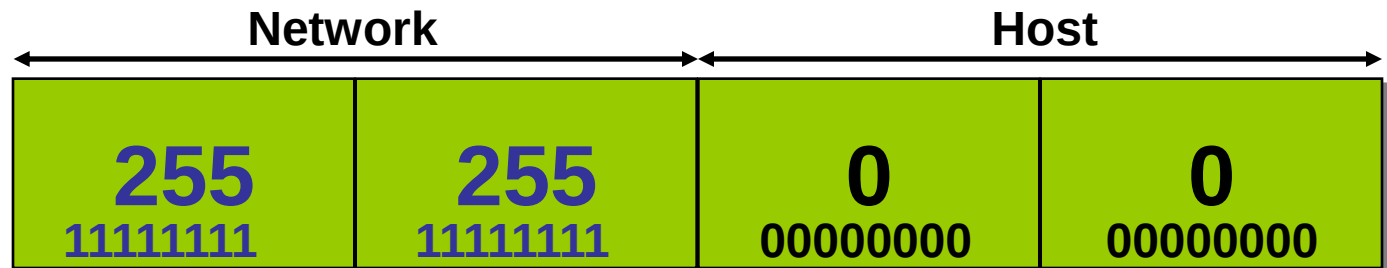


Subnet mask

IP
Address

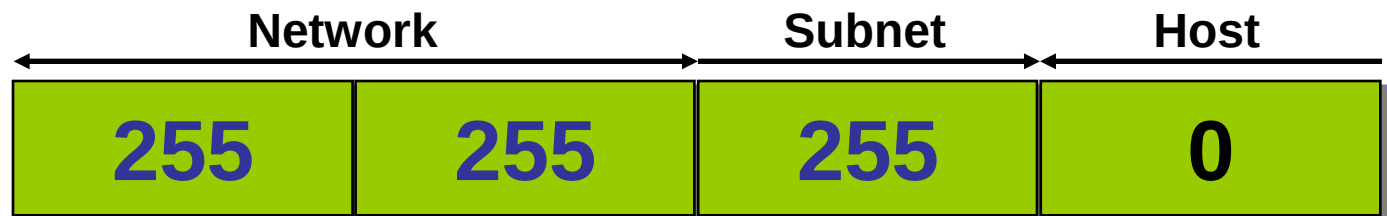


Default
Subnet
Mask

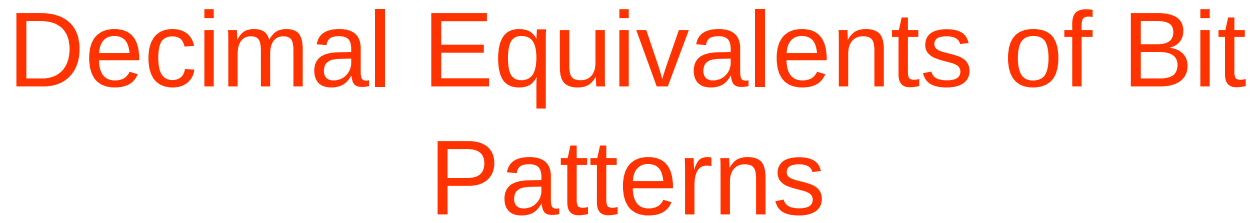


Also written as “/16” where 16 represents the number of 1s in the mask.

Borrow 8-bit
Subnet
Mask



Also written as “/24” where 24 represents the number of 1s in the mask.

[illegible]



Subnet Mask without Subnets

	Network		Host	
172.16.2.160	10101100	00010000	00000010	10100000
255.255.0.0	11111111	11111111	00000000	00000000
	10101100	00010000	00000000	00000000
Network Address	172	16	0	0

use the default subnet mask



Subnet Mask with Subnets

	Network		Subnet	Host
172.16.2.160	10101100	00010000	00000010	10100000
255.255.255.0	11111111	11111111	11111111	00000000
	10101100	00010000	128 192 224 240 248 252 254 255	00000000
Network Address	172	16	2	0

Network number extended by eight bits



Subnet Mask with Subnets

	Network		Subnet	Host
172.16.2.160	10101100	00010000	00000010	10100000
255.255.255.192	11111111	11111111	11111111	11000000
			128 192 224 240 248 252 254 255	128 192 224 240 248 252 254 255
	10101100	00010000	00000010	10000000
Network Address	172	16	2	128

Network number extended by ten bits



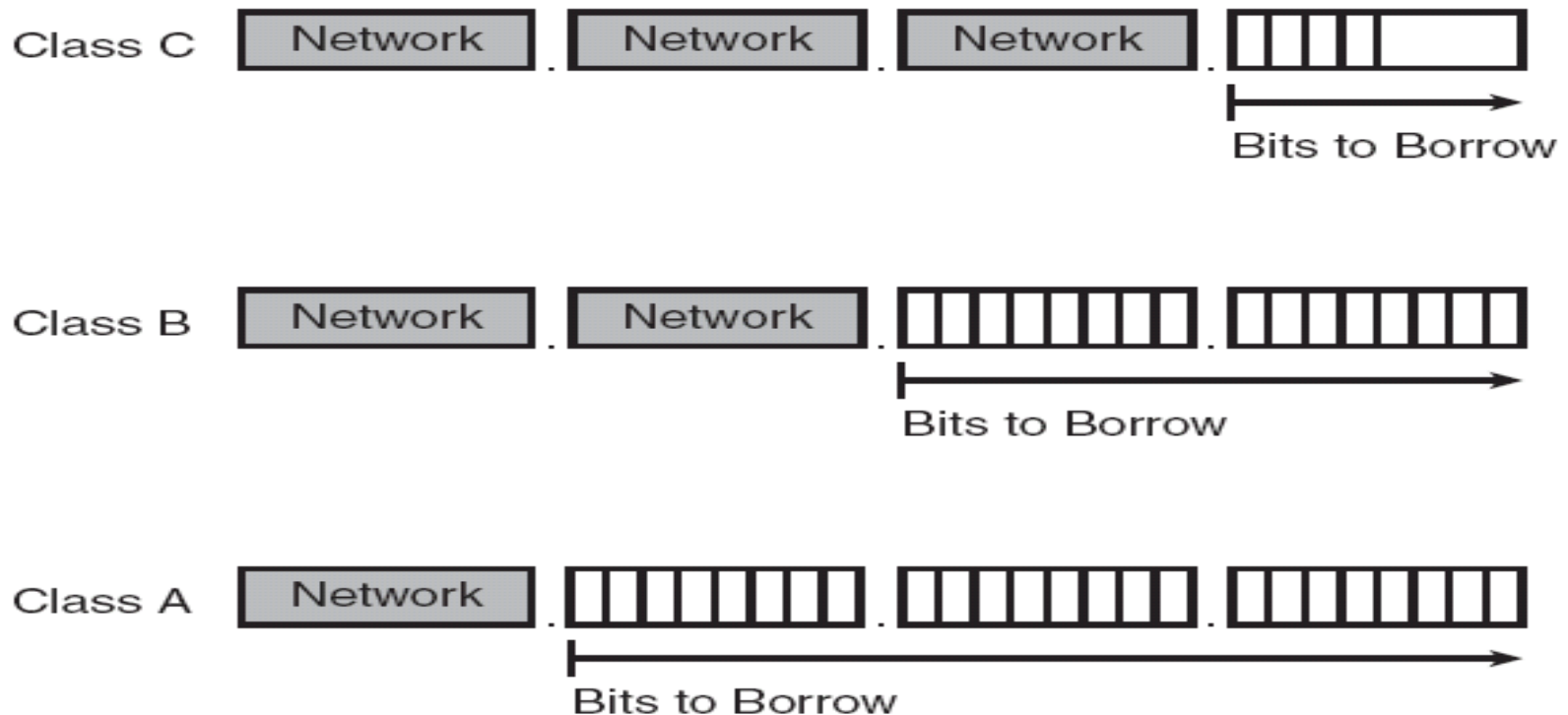
Exercise

172	16	2	160
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172.16.2.160	10101100	00010000	00000010	10100000	Host
255.255.255.192	11111111	11111111	11111111	11000000	Mask
	10101100	00010000	00000010	10000000	Subnet
	172	16	2	128	
	10101100	00010000	00000010	10111111	Broadcast
	172	16	2	191	
	10101100	00010000	00000010	10000001	First device
	172	16	2	129	
	10101100	00010000	00000010	10111110	Last device
	172	16	2	190	



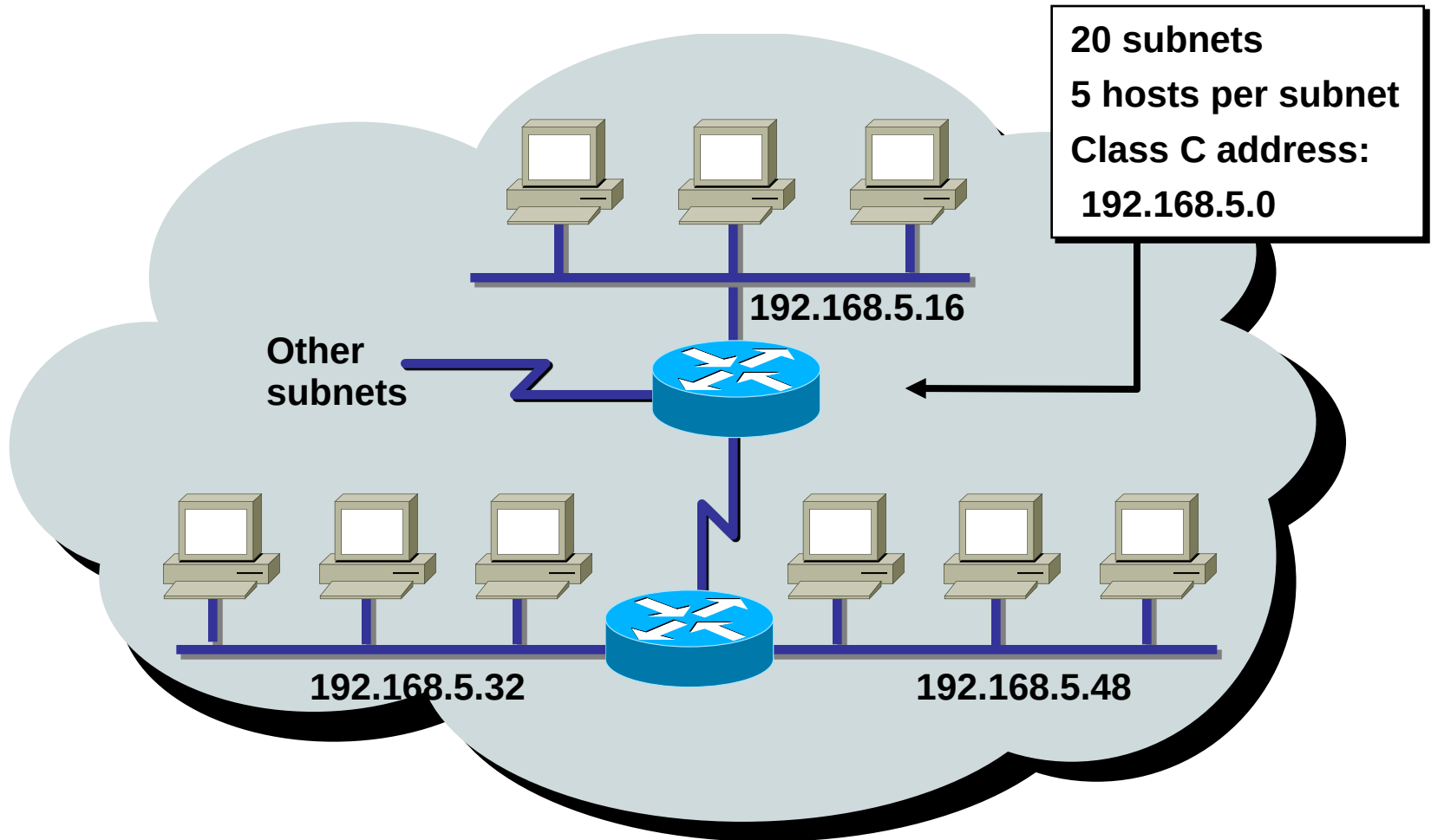
Review of subnets



- No subnet → default subnet mask

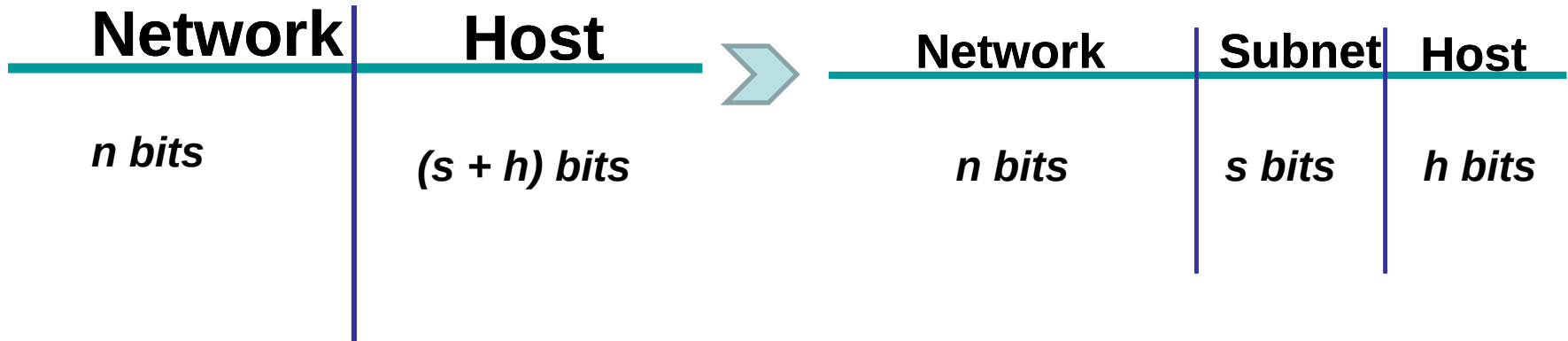


Subnet planning





Subnet planning



- $(n + s + h) = 32$ bits
- Number of subnets = 2^s
- Number of host/subnet = $2^h - 2$ (**subnet address** & **broadcast address**)

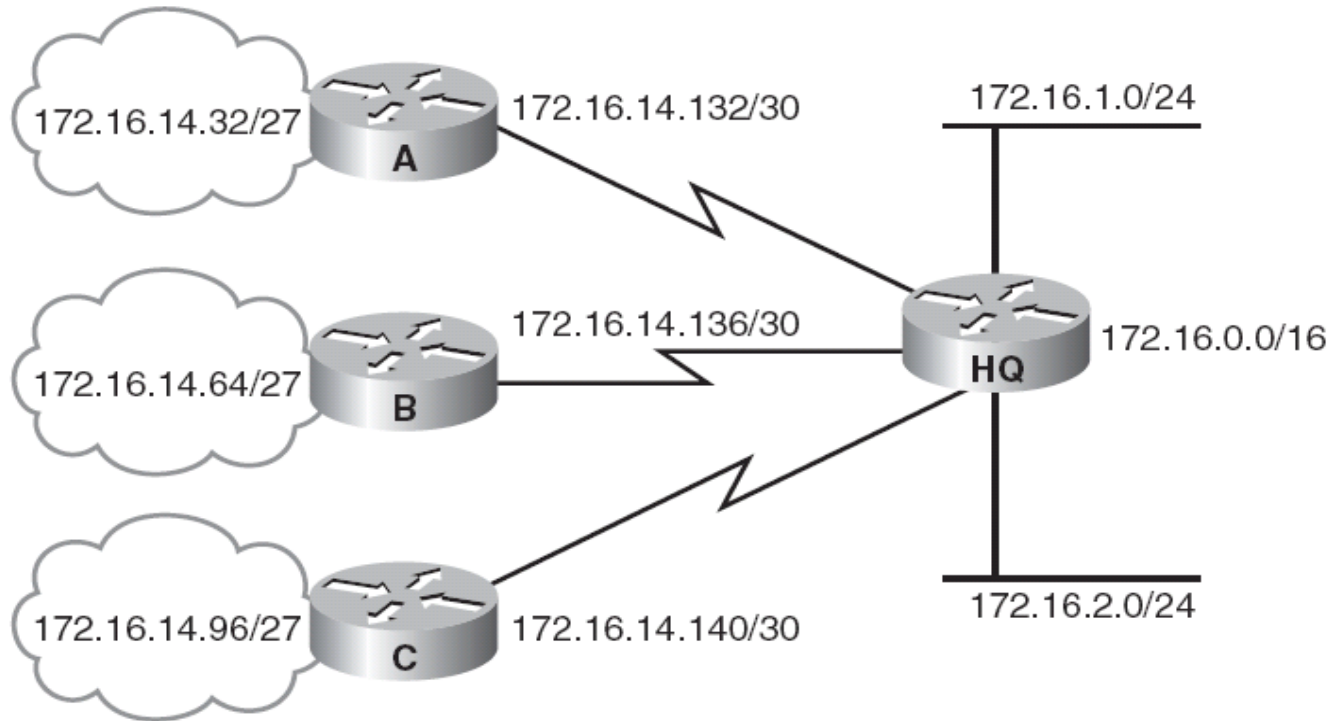


Subnet planning

- Example: 172.16.0.0/16, 100 subnets, max number of host
 - How many bits will need to be borrowed?
 - What is the new subnet mask?
 - What are the first four subnets?
 - What are the ranges of host addresses for the four subnets?



VLSM (Variable Length Subnet Mask)





VLSM (Variable Length Subnet Mask)



-
- More than 1 subnet mask in a network
 - Subnet an already subnetted address.
 - Benefits
 - More efficient use of IP addresses (i.e. only two hosts to be used on the WAN links)
 - Greater capability to use route summarization
 - Only subnets that are unused can be further subnetted.



VLSM (Variable Length Subnet Mask)

Subnetted Address: 172.16.32.0/20

In Binary 10101100.00010000.00100000.00000000

VLSM Address: 172.16.32.0/26

In Binary 10101100.00010000.00100000.00000000

1st subnet:	172	.	16	.0010	0000.00	000000 = 172.16.32.0/26
2nd subnet:	172	.	16	.0010	0000.01	000000 = 172.16.32.64/26
3rd subnet:	172	.	16	.0010	0000.10	000000 = 172.16.32.128/26
4th subnet:	172	.	16	.0010	0000.11	000000 = 172.16.32.192/26
5th subnet:	172	.	16	.0010	001.00	000000 = 172.16.33.0/26
	Network			Subnet	VLSM Subnet	Host